

Plexiphon-inspired Spatial Effect — Design Brief

Project context

Standalone DSP effect for the ER-301 successor platform. **Not a traditional reverb** — framed as a “macro spatial simulation” / topology-morphing space effect. Aesthetic target: glitchy and shapeable on one end, lush on the other, with continuous travel between.

Reference points

- **Make Noise Plexiphon** (Tom Erbe, 2026): single “Plexus” parameter modulates both the *number* of active feedback paths and their *entanglement*; Size sets temporal relation between paths (delay-time at one extreme, room-size at the other). No mode switching — same network being structurally reconfigured.
- **Intellijel Rainmaker**: 64-tap comb with addressable per-tap state. Valued for whipsaw between states and tap-as-event identity.
- **Erbe-Verb**: continuous parameterization across topology classes via a single perceptual axis.

Constraints

- **Target SoC**: AM3358 Sitara (Cortex-A8, NEON). Modest budget. Rules out convolution, dense allpass tanks, waveguide meshes, large-N full FDNs. Sparse matrices, block-diagonal structures, short modulated delays only.
- **Macro-controlled UI**: a few perceptually meaningful knobs, not per-tap editing. Tap/reflector pattern is *generated* from macro params (geometry + seed + distribution), not user-edited.
- **Aesthetic**: glitchy/addressable at low density, textural at high density. Distinct events should remain perceptible at the sparse end.

Reverb topology landscape (for orientation)

Classical families: Schroeder (parallel combs + series allpass), FDN/Jot (N delays + unitary mixing matrix + per-line absorption), nested allpass (Dattorro/Gardner), convolution, waveguide mesh, SDN, modal, velvet noise, spectral/granular.

Less-explored, fertile ground for novelty: time-varying mixing matrices, sparse/topology-morphing FDNs, nonlinear feedback, modulated allpass *coefficients* (not just delay length), reservoir-style random recurrent networks, energy-conserving lossless prototypes with shaped loss, fractal/self-similar delay sets, cross-modal coupling (pitch/freq shift in loop).

The framework: pick a graph topology, decide lossless prototype + where loss enters, pick what's modulatable in real time, then find *one perceptual axis* that exposes the structure. Good algorithms are clever parameterizations of clever DSP.

Three candidate topologies (ranked by fit, all to be prototyped)

1. Selective-feedback multi-tap (Rainmaker++) $N=32-64$ taps, each with (delay, gain, sign, pan, optional 1-pole). A recycle stage selects k of N taps to sum back into the write head with sparse weighting. Glitchy at small k with rhythmic spacing; lush as k grows and spacing densifies. Tap pattern selectable: rhythmic / stochastic-velvet / geometric. The Plexiphon morph axis is "tap density $\times$ feedback connectivity."

2. Micro-FDN cluster lattice M small FDNs (each $N=2-4$, Householder mixing) sparsely cross-coupled. Each cluster reads as a discrete reflector group at low connectivity, blurs to reverb at high. Morph param raises both M -active and inter-cluster coupling. Each cluster small enough that low-connectivity output reads as events, not tail.

3. Virtual reflector field (most distinctive, best fit for "macro spatial simulation") N point reflectors in a 2D virtual space; each is a tap with delay $\propto$ distance, gain $\propto 1/r$, pan from azimuth. A listener-motion parameter translates/rotates through the field, producing Doppler (delay-line modulation) and panning shifts as a coherent gesture. Sparse field = glitchy and spatial; dense field with motion = textural cloud. Doesn't sound like standard reverb because the structural assumption is geometric, not statistical.

Working hypothesis (to validate via prototype)

Hybrid of (1) and (3): multi-tap field where tap pattern is *generated* from a virtual geometry, with sparse selectable feedback recycling. Rainmaker-style whipsaw from individually addressable reflectors. Coherent spatial gestures from listener motion moving all taps consistently. Cheap implementation: under the hood it's a multi-tap delay with a parameter generator on top.

CPU notes

64 taps $\times$ (read, gain, pan, 1-pole) at 48 kHz with NEON is well under 1% of an A8. Cost centers to watch: large mixing matrices (keep sparse/block-diagonal), modulation on long delays (cache misses — modulate short delays only), high-order allpass chains.

Glitch ↔ lush axis — implementation options, cheap to expensive

- Per-tap or post-bus jitter (small randomized delay perturbation) — cheapest
- One short modulated allpass per tap (~4 ops/tap)
- Topology-density parameter: active tap count + connectivity — heaviest but most musical, changes the *kind* of texture rather than blurring a fixed one

Recommendation: primary control on density + connectivity; reserve smear/allpass-jitter as a secondary “soften” knob.

Open questions

- Does final decision require A/B'ing prototypes of all three topologies, or can (3) be skipped if hybrid (1+3) is built directly?
- Macro parameter set TBD until prototypes exist — likely some combination of: density, connectivity, size, decay, soften, motion, seed.
- Stereo strategy: dual-mono vs. coupled cross-feed (Plexiphon-style Couple/Skew) — defer until mono behavior is settled.

Next step for dev agent

Build minimal prototypes of (1), (2), (3) in the ER-301 emulator environment with placeholder UI. Goal is audible comparison, not optimization. Decision on which to develop further is contingent on listening.